

13.6 DIRECTIONAL DERIVATIVES

The partial derivatives $f_x(x, y)$ and $f_y(x, y)$ represent the rates of change of $f(x, y)$ in directions parallel to the x - and y -axes. In this section we will investigate rates of change of $f(x, y)$ in other directions.

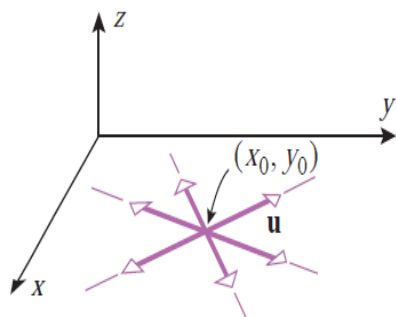
Suppose that we wish to compute the instantaneous rate of change of a function $f(x, y)$ with respect to distance from a point (x_0, y_0) in some direction. Since there are infinitely many different directions from (x_0, y_0) in which we could move, we need a convenient method for describing a specific direction starting at (x_0, y_0) . One way to do this is to use a unit vector

$$\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j}$$

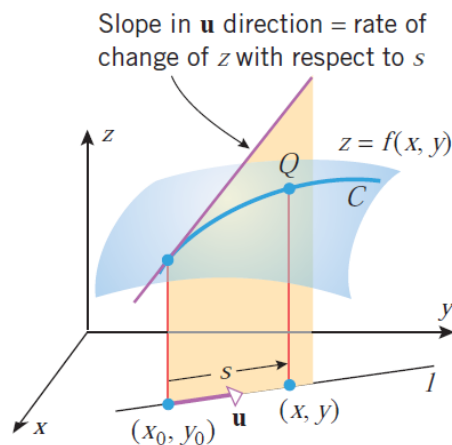
that has its initial point at (x_0, y_0) and points in the desired direction (Figure 13.6.1). This vector determines a line l in the xy -plane that can be expressed parametrically as

$$x = x_0 + su_1, \quad y = y_0 + su_2 \quad (1)$$

Since $\mathbf{u}$ is a unit vector, s is the arc length parameter that has its reference point at (x_0, y_0) and has positive values in the direction of $\mathbf{u}$.



▲ Figure 13.6.1



▲ Figure 13.6.2

13.6.1 DEFINITION If $f(x, y)$ is a function of x and y , and if $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j}$ is a unit vector, then the **directional derivative of f in the direction of $\mathbf{u}$** at (x_0, y_0) is denoted by $D_{\mathbf{u}}f(x_0, y_0)$ and is defined by

$$D_{\mathbf{u}}f(x_0, y_0) = \frac{d}{ds} [f(x_0 + su_1, y_0 + su_2)]_{s=0} \quad (2)$$

provided this derivative exists.

Geometrical Interpretations:

Geometrically, $D_{\mathbf{u}}f(x_0, y_0)$ can be interpreted as the *slope of the surface $z = f(x, y)$ in the direction of $\mathbf{u}$* at the point $(x_0, y_0, f(x_0, y_0))$ (Figure 13.6.2). Usually the value of $D_{\mathbf{u}}f(x_0, y_0)$ will depend on both the point (x_0, y_0) and the direction $\mathbf{u}$. Thus, at a fixed point the slope of the surface may vary with the direction (Figure 13.6.3). Analytically, the directional derivative represents the *instantaneous rate of change of $f(x, y)$ with respect to distance in the direction of $\mathbf{u}$* at the point (x_0, y_0) .

► **Example 1** Let $f(x, y) = xy$. Find and interpret $D_{\mathbf{u}}f(1, 2)$ for the unit vector

$$\mathbf{u} = \frac{\sqrt{3}}{2}\mathbf{i} + \frac{1}{2}\mathbf{j}$$

Solution. It follows from Equation (2) that

$$D_{\mathbf{u}}f(1, 2) = \frac{d}{ds} \left[f \left(1 + \frac{\sqrt{3}s}{2}, 2 + \frac{s}{2} \right) \right]_{s=0}$$

Since

$$f \left(1 + \frac{\sqrt{3}s}{2}, 2 + \frac{s}{2} \right) = \left(1 + \frac{\sqrt{3}s}{2} \right) \left(2 + \frac{s}{2} \right) = \frac{\sqrt{3}}{4}s^2 + \left(\frac{1}{2} + \sqrt{3} \right)s + 2$$

we have

$$\begin{aligned} D_{\mathbf{u}}f(1, 2) &= \frac{d}{ds} \left[\frac{\sqrt{3}}{4}s^2 + \left(\frac{1}{2} + \sqrt{3} \right)s + 2 \right]_{s=0} \\ &= \left[\frac{\sqrt{3}}{2}s + \frac{1}{2} + \sqrt{3} \right]_{s=0} = \frac{1}{2} + \sqrt{3} \end{aligned}$$

Since $\frac{1}{2} + \sqrt{3} \approx 2.23$, we conclude that if we move a small distance from the point $(1, 2)$ in the direction of $\mathbf{u}$, the function $f(x, y) = xy$ will increase by about 2.23 times the distance moved. ◀

13.6.2 DEFINITION If $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j} + u_3\mathbf{k}$ is a unit vector, and if $f(x, y, z)$ is a function of x , y , and z , then the *directional derivative of f in the direction of $\mathbf{u}$* at (x_0, y_0, z_0) is denoted by $D_{\mathbf{u}}f(x_0, y_0, z_0)$ and is defined by

$$D_{\mathbf{u}}f(x_0, y_0, z_0) = \frac{d}{ds} [f(x_0 + su_1, y_0 + su_2, z_0 + su_3)]_{s=0} \quad (3)$$

provided this derivative exists.

13.6.3 THEOREM

(a) If $f(x, y)$ is differentiable at (x_0, y_0) , and if $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j}$ is a unit vector, then the directional derivative $D_{\mathbf{u}}f(x_0, y_0)$ exists and is given by

$$D_{\mathbf{u}}f(x_0, y_0) = f_x(x_0, y_0)u_1 + f_y(x_0, y_0)u_2 \quad (4)$$

(b) If $f(x, y, z)$ is differentiable at (x_0, y_0, z_0) , and if $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j} + u_3\mathbf{k}$ is a unit vector, then the directional derivative $D_{\mathbf{u}}f(x_0, y_0, z_0)$ exists and is given by

$$D_{\mathbf{u}}f(x_0, y_0, z_0) = f_x(x_0, y_0, z_0)u_1 + f_y(x_0, y_0, z_0)u_2 + f_z(x_0, y_0, z_0)u_3 \quad (5)$$

Note-1:

A unit vector $\mathbf{u}$ in the xy -plane can also be expressed in the following form

$$\mathbf{u} = \cos \phi \mathbf{i} + \sin \phi \mathbf{j} \quad (6)$$

where ϕ is the angle from the positive x -axis to $\mathbf{u}$. Thus, Formula (4) can also be expressed as

$$D_{\mathbf{u}}f(x_0, y_0) = f_x(x_0, y_0) \cos \phi + f_y(x_0, y_0) \sin \phi \quad (7)$$

► **Example 2** Find the directional derivative of $f(x, y) = e^{xy}$ at $(-2, 0)$ in the direction of the unit vector that makes an angle of $\pi/3$ with the positive x -axis.

Solution. The partial derivatives of f are

$$\begin{aligned} f_x(x, y) &= ye^{xy}, & f_y(x, y) &= xe^{xy} \\ f_x(-2, 0) &= 0, & f_y(-2, 0) &= -2 \end{aligned}$$

The unit vector $\mathbf{u}$ that makes an angle of $\pi/3$ with the positive x -axis is

$$\mathbf{u} = \cos(\pi/3)\mathbf{i} + \sin(\pi/3)\mathbf{j} = \frac{1}{2}\mathbf{i} + \frac{\sqrt{3}}{2}\mathbf{j}$$

Thus, from (7)

$$\begin{aligned} D_{\mathbf{u}}f(-2, 0) &= f_x(-2, 0) \cos(\pi/3) + f_y(-2, 0) \sin(\pi/3) \\ &= 0(1/2) + (-2)(\sqrt{3}/2) = -\sqrt{3} \quad \blacktriangleleft \end{aligned}$$

► **Example 3** Find the directional derivative of $f(x, y, z) = x^2y - yz^3 + z$ at the point $(1, -2, 0)$ in the direction of the vector $\mathbf{a} = 2\mathbf{i} + \mathbf{j} - 2\mathbf{k}$.

Solution. The partial derivatives of f are

$$\begin{aligned} f_x(x, y, z) &= 2xy, & f_y(x, y, z) &= x^2 - z^3, & f_z(x, y, z) &= -3yz^2 + 1 \\ f_x(1, -2, 0) &= -4, & f_y(1, -2, 0) &= 1, & f_z(1, -2, 0) &= 1 \end{aligned}$$

Since $\mathbf{a}$ is not a unit vector, we normalize it, getting

$$\mathbf{u} = \frac{\mathbf{a}}{\|\mathbf{a}\|} = \frac{1}{\sqrt{9}}(2\mathbf{i} + \mathbf{j} - 2\mathbf{k}) = \frac{2}{3}\mathbf{i} + \frac{1}{3}\mathbf{j} - \frac{2}{3}\mathbf{k}$$

Formula (5) then yields

$$D_{\mathbf{u}}f(1, -2, 0) = (-4)\left(\frac{2}{3}\right) + \frac{1}{3} - \frac{2}{3} = -3 \quad \blacktriangleleft$$

Example 1 Find each of the directional derivatives.

- (a) $D_{\vec{u}}f(2, 0)$ where $f(x, y) = xe^{xy} + y$ and $\vec{u}$ is the unit vector in the direction of $\theta = \frac{2\pi}{3}$.
- (b) $D_{\vec{u}}f(x, y, z)$ where $f(x, y, z) = x^2z + y^3z^2 - xyz$ in the direction of $\vec{v} = \langle -1, 0, 3 \rangle$.

Solution

- (a) $D_{\vec{u}}f(2, 0)$ where $f(x, y) = xe^{xy} + y$ and $\vec{u}$ is the unit vector in the direction of $\theta = \frac{2\pi}{3}$.

We'll first find $D_{\vec{u}}f(x, y)$ and then use this a formula for finding $D_{\vec{u}}f(2, 0)$. The unit vector giving the direction is,

$$\vec{u} = \left\langle \cos\left(\frac{2\pi}{3}\right), \sin\left(\frac{2\pi}{3}\right) \right\rangle = \left\langle -\frac{1}{2}, \frac{\sqrt{3}}{2} \right\rangle$$

So, the directional derivative is,

$$D_{\vec{u}}f(x, y) = \left(-\frac{1}{2}\right)(e^{xy} + xye^{xy}) + \left(\frac{\sqrt{3}}{2}\right)(x^2e^{xy} + 1)$$

Now, plugging in the point in question gives,

$$D_{\vec{u}}f(2, 0) = \left(-\frac{1}{2}\right)(1) + \left(\frac{\sqrt{3}}{2}\right)(5) = \frac{5\sqrt{3}-1}{2}$$

(b) $D_{\vec{u}}f(x, y, z)$ where $f(x, y, z) = x^2z + y^3z^2 - xyz$ in the direction of $\vec{v} = \langle -1, 0, 3 \rangle$.

In this case let's first check to see if the direction vector is a unit vector or not and if it isn't convert it into one. To do this all we need to do is compute its magnitude.

$$\|\vec{v}\| = \sqrt{1 + 0 + 9} = \sqrt{10} \neq 1$$

So, it's not a unit vector. Recall that we can convert any vector into a unit vector that points in the same direction by dividing the vector by its magnitude. So, the unit vector that we need is,

$$\vec{u} = \frac{1}{\sqrt{10}} \langle -1, 0, 3 \rangle = \left\langle -\frac{1}{\sqrt{10}}, 0, \frac{3}{\sqrt{10}} \right\rangle$$

The directional derivative is then,

$$\begin{aligned} D_{\vec{u}}f(x, y, z) &= \left(-\frac{1}{\sqrt{10}} \right) (2xz - yz) + (0) (3y^2z^2 - xz) + \left(\frac{3}{\sqrt{10}} \right) (x^2 + 2y^3z - xy) \\ &= \frac{1}{\sqrt{10}} (3x^2 + 6y^3z - 3xy - 2xz + yz) \end{aligned}$$

■ THE GRADIENT

Formula (4) can be expressed in the form of a dot product as

$$\begin{aligned} D_{\mathbf{u}}f(x_0, y_0) &= (f_x(x_0, y_0)\mathbf{i} + f_y(x_0, y_0)\mathbf{j}) \cdot (u_1\mathbf{i} + u_2\mathbf{j}) \\ &= (f_x(x_0, y_0)\mathbf{i} + f_y(x_0, y_0)\mathbf{j}) \cdot \mathbf{u} \end{aligned}$$

Similarly, Formula (5) can be expressed as

$$D_{\mathbf{u}}f(x_0, y_0, z_0) = (f_x(x_0, y_0, z_0)\mathbf{i} + f_y(x_0, y_0, z_0)\mathbf{j} + f_z(x_0, y_0, z_0)\mathbf{k}) \cdot \mathbf{u}$$

13.6.4 DEFINITION

(a) If f is a function of x and y , then the **gradient of f** is defined by

$$\nabla f(x, y) = f_x(x, y)\mathbf{i} + f_y(x, y)\mathbf{j} \quad (8)$$

(b) If f is a function of x , y , and z , then the **gradient of f** is defined by

$$\nabla f(x, y, z) = f_x(x, y, z)\mathbf{i} + f_y(x, y, z)\mathbf{j} + f_z(x, y, z)\mathbf{k} \quad (9)$$

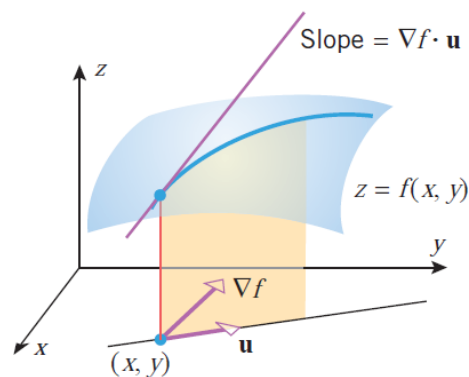
Formulas (4) and (5) can now be written as

$$D_{\mathbf{u}}f(x_0, y_0) = \nabla f(x_0, y_0) \cdot \mathbf{u} \quad (10)$$

and

$$D_{\mathbf{u}}f(x_0, y_0, z_0) = \nabla f(x_0, y_0, z_0) \cdot \mathbf{u} \quad (11)$$

Formula (10) can be interpreted to mean that the slope of the surface $z = f(x, y)$ at the point (x_0, y_0) in the direction of $\mathbf{u}$ is the dot product of the gradient with $\mathbf{u}$ (Figure 13.6.4).



▲ Figure 13.6.4

Example 2 Find each of the directional derivative.

- (a) $D_{\vec{u}}f(\vec{x})$ for $f(x, y) = x \cos(y)$ in the direction of $\vec{v} = \langle 2, 1 \rangle$.
 (b) $D_{\vec{u}}f(\vec{x})$ for $f(x, y, z) = \sin(yz) + \ln(x^2)$ at $(1, 1, \pi)$ in the direction of $\vec{v} = \langle 1, 1, -1 \rangle$.

Solution

(a) $D_{\vec{u}}f(\vec{x})$ for $f(x, y) = x \cos(y)$ in the direction of $\vec{v} = \langle 2, 1 \rangle$.

Let's first compute the gradient for this function.

$$\nabla f = \langle \cos(y), -x \sin(y) \rangle$$

Also, as we saw earlier in this section the unit vector for this direction is,

$$\vec{u} = \left\langle \frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}} \right\rangle$$

The directional derivative is then,

$$\begin{aligned} D_{\vec{u}}f(\vec{x}) &= \langle \cos(y), -x \sin(y) \rangle \cdot \left\langle \frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}} \right\rangle \\ &= \frac{1}{\sqrt{5}} (2 \cos(y) - x \sin(y)) \end{aligned}$$

(b) $D_{\vec{u}}f(\vec{x})$ for $f(x, y, z) = \sin(yz) + \ln(x^2)$ at $(1, 1, \pi)$ in the direction of $\vec{v} = \langle 1, 1, -1 \rangle$.

In this case are asking for the directional derivative at a particular point. To do this we will first compute the gradient, evaluate it at the point in question and then do the dot product. So, let's get the gradient.

$$\begin{aligned}\nabla f(x, y, z) &= \left\langle \frac{2}{x}, z \cos(yz), y \cos(yz) \right\rangle \\ \nabla f(1, 1, \pi) &= \left\langle \frac{2}{1}, \pi \cos(\pi), \cos(\pi) \right\rangle = \langle 2, -\pi, -1 \rangle\end{aligned}$$

Next, we need the unit vector for the direction,

$$\|\vec{v}\| = \sqrt{3} \qquad \vec{u} = \left\langle \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}} \right\rangle$$

Finally, the directional derivative at the point in question is,

$$\begin{aligned}D_{\vec{u}}f(1, 1, \pi) &= \langle 2, -\pi, -1 \rangle \cdot \left\langle \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}} \right\rangle \\ &= \frac{1}{\sqrt{3}}(2 - \pi + 1) \\ &= \frac{3 - \pi}{\sqrt{3}}\end{aligned}$$

Practice Problem:

Chapter-13.6:

Problems: 1,3,5,7, 8, 11,12,13,16-18,22-26,33,35,42,44

Thank you for your attention!